

Composition of Functions

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$ # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(2)$.	ANSWER: 24 # _____ Given $f(x) = 2x + 3$, find $(f \circ f)(1)$.
ANSWER: 8 # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(0)$.	ANSWER: $2x^2 + 1$ # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, write the expression for $(g \circ f)(x)$.

ANSWER: 10 # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(g \circ f)(1)$.	ANSWER: 9 # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(g \circ f)(0)$.
ANSWER: 0 # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, write the expression for $(f \circ g)(x)$.	ANSWER: $4x^2 + 12x + 8$ # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, state the domain of $\left(\frac{f}{g}\right)(x)$ in interval notation.
ANSWER: 1 # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f + g)(2)$.	ANSWER: 13 # _____ Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \cdot g)(1)$.